Virtual Machine Management

What is a Virtual Machine?

A **Virtual Machine (VM)** is an emulated computer system running on top of physical hardware using a **hypervisor**. It includes its own OS, memory, disk, and CPU.

Types of Hypervisors

Type	Description	Examples
Type 1	Bare-metal; runs directly on hardware	VMware ESXi, Microsoft Hyper-V, KVM
Type 2	Hosted; runs over an existing OS	VirtualBox, VMware Workstation

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VM Lifecycle Operations

Operation	Description
Create	Define a new VM with OS, CPU, RAM, disk
Start/Stop	Boot or shut down the VM
Pause/Resume	Freeze and later continue the VM
Snapshot	Capture the current state for rollback
Clone	Duplicate a VM
Delete	Remove the VM definition and storage

VM Tools & Interfaces

Tool	Usage
Virt-Manager	GUI to manage KVM-based VMs

virsh	CLI tool for KVM/QEMU management
vSphere	GUI for VMware ESXi
VBoxManage	CLI for VirtualBox management

🔹 VM Resource Allocation

- **CPU & vCPU**: Assign one or more logical CPUs
- **RAM**: Allocate virtual memory to the guest OS
- **Disk**: Use virtual disk images (VDI, VMDK, QCOW2)
- **GPU Pass-through**: Allows VM to use host GPU directly

🔐 Security Best Practices

- Use **separate user accounts** for VM management
- Isolate VM traffic using **virtual firewalls**
- Enable **virtual TPM**, **UEFI Secure Boot**, and **guest isolation**
- Regularly **update guest OS** and **hypervisor patches**

🌐 Virtual Machine Network Configuration

🔌 VM Network Modes

Mode	Description	Use Case
NAT (Network Address Translation)	VM shares host IP, gets internet	Default for VirtualBox
Bridged	VM acts like a separate device on the LAN	Best for services
Host-Only	VM communicates only with host	Lab/test environment
Internal	VM-to-VM communication only	Isolated virtual LAN
Custom Networks	Using VLANs or SDN	Advanced scenarios

Bridged vs NAT

Feature	NAT	Bridged
Internet Access	Yes	Yes
LAN Access	No	Yes
Port Forwarding Required?	Yes	No
Public IP?	No	Yes

Tools for Configuring Networking

Tool/Command	Use
nmcli, ip, ifconfig	Configure IP settings
VirtualBox GUI / VBoxManage	Change NIC modes
VMware vSphere / Workstation	Set virtual switch types
libvirt (virsh/net-* commands)	Manage virtual networks in KVM

Network Security

- Use **virtual firewalls** (e.g., iptables/nftables)
- Apply **MAC address filtering** and **IP whitelisting**
- Segment traffic using **VLANs**
- Disable unused NICs and services inside the VM

Advanced Networking

- **SR-IOV**: Direct hardware access to VM NICs
- **vSwitches**: Software-based switches inside hypervisors
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VXLAN: Virtual LAN across Layer 3 (used in OpenStack/Kubernetes)